

Interaction

Now we are going to discuss the 5 concepts that are the backbone of interaction design:

- 1) Affordance
- 2) Conceptual Model
- 3) Natural Mapping
- 4) Feed back
- 5) Constraints

1) Affordance:

"Does the user know what actions can be done"

An **affordance** is a property of an object that shows what actions are possible. It determines how an object can be used.

day/date

e.g.

If a door has a handle, it shows pull.

• A flat plate on door shows push.

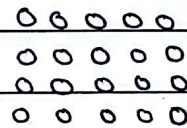
• A 3D styled button affords clicking

• A slider thumb affords dragging

We can use Gestalt principles to help create & reinforce perceived affordances

Gestalt Laws:

1) Proximity:

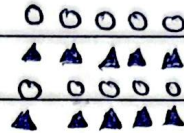


Elements positioned close to each other are perceived as a single group or unit.

Bereh: Group Interactive controls
together to signify unified function.

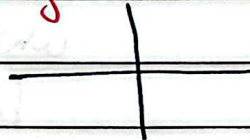
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2) Similarity:



• Elements that look similar are processed to share same function or meaning

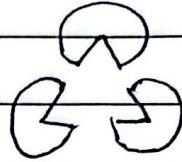
3) Continuity:



The human eye naturally follows a path, a line or curve.

eg A row of image cards in websites with one image partially cut off on screen ~~shows~~ affords swiping horizontally.

4) Closure:



The mind automatically fills in missing visual information to perceive a complete shape.

- Use minimal visual elements while maintaining clear functional affordances.

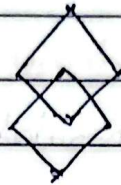
~~5) Figure / Ground:~~

5) Area:



Elements in same boundary are perceived as belonging together.

c) Symmetry



visual elements that are arranged symmetrically around a central axis are perceived as having equal importance.

2) Conceptual Model:

A **Conceptual Model** is a high level mental stimulation formed by the user about how a system works.

"Does the user understand how the system operates?"

E.g. The concept of Desktop, Folders & Trash can on an OS is a conceptual model designed to mimic a real physical office desk.

3) Natural Mapping:

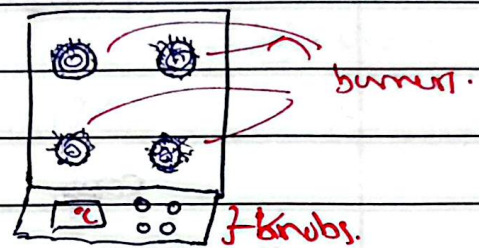
Mappings is the relationship between spatial controls & their results in the physical or digital world.

"Is the relationship between control & effect obvious?"

Example:

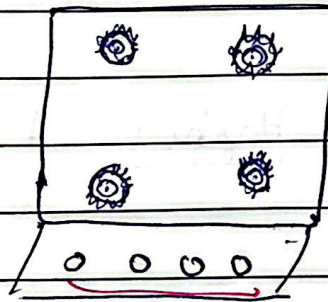
Good Mapping:

Arranging 4 stove-top knobs in the exact same spatial layout as the 4 burners.



Bad Mapping:

Arrange the 4 knobs in different layout



now user needs to guess which knob is of which burner

4) Feedback:

It is the sending information back to user about what actions has been performed.

Feedback must be: **immediate, clear & unambiguous.**

"Does user know what system did?"

Example:

visual: A button changing color when ~~click~~ pressed.

Auditory: A click sound when locking phone.

Haptic: A vibration when you tap on screen.

5) Constraints:

Constraints limits the set of possible actions a user can perform at any given moment

"Is the user restricted from making any critical errors"

Example:

- You can only insert an SD-card or USB-C cable in specific ways.

- Cultural constraints: Red = Stop, green = go.

A few evaluation metrics:

Now we will see a few evaluation metrics that can be used to check the interactivity & ease of use of anything:

1) Fitt's law:

It predicts the time required to rapidly move to a target area as a function of target's distance (D) & its width (w).

$$MT = a + b \cdot \log_2 \left(1 + \frac{D}{w} \right)$$

MT = movement Time

a, b = constants.

D = distance

w = width } of target.

• increasing w reduces movement time

2) Hick Hyman's Law (Hick's Law)

This tells the ~~time to make~~ ^{time to make} ~~number of~~ ^{of} decisions when "n" number of choices are present.

$$T = a + b \cdot \log_2(n+1)$$

n = equal probability choices.

b = constant.

3) Error Rate:

$$\text{Error rate per task} = \frac{\text{Total errors across all users}}{\text{Total task Attempts}}$$

4) Task completion Rate.

$$\text{Completion Rate} = \frac{\text{Successfully completed tasks}}{\text{Total tasks}} \times 100$$

5) Throughput.

This measures the overall information processing capacity of a system.

$$TP = \frac{IDe}{MT}$$

MT = movement time from Fitts law.

$$IDe = \log_2 \left(\frac{D}{w_e} + 1 \right)$$

D = Distance

w_e = Effective width.